Research Topics Draft 1

Set 1: Fraud Mechanics & Transaction Structuring

- How do cybercrime syndicates structure multi-hop transfers to balance network gas fees against analytical detection difficulty?
- What behavioral differences distinguish a decentralized liquidity provider's smart contract calls from automated peeling scripts?
- Why do high-volume scam networks preferentially deploy on specific networks (e.g., TRON vs. EVM vs. Bitcoin) based on throughput and fee economics?
- How does token standard behavior (e.g., ERC-20/TRC-20 transferFrom vs. native coin transfers) impact the visibility of layered assets?
- How do nesting services and sub-account architectures on major exchanges obscure the ultimate beneficiary's identity?

Set 2: Graph Theory & Traversal Optimization

- What algorithmic pruning strategies prevent combinatorial explosion during Breadth-First Search (BFS) on high-degree contract nodes?
- How can graph centrality metrics (Betweenness, PageRank, Degree Centrality) identify money-mule aggregators among thousands of burner wallets?
- What edge-weighting formulations best capture the decay of transaction taint over multi-hop splits and partial consolidations?
- How does graph traversal complexity compare when querying directed acyclic graphs (DAGs) on UTXO models versus state transitions on account models?
- How can bidirectional search (forward from victim, backward from known VASP deposit addresses) optimize traversal speed?

Set 3: Address Clustering & Entity Resolution

- What mathematical and structural rules reliably distinguish a change address from an intermediary forwarding address?
- How do gas-funding heuristics (tracing parent funding wallets) link disparate non-custodial addresses to a single operator?
- What behavioral markers differentiate a single entity's multi-wallet cluster from a large exchange's shared hot-wallet pool?
- How does the common-input-ownership heuristic degrade in accuracy when encountering CoinJoin or privacy-enhanced UTXO transactions?
- What confidence-scoring metrics evaluate the probabilistic certainty of an address cluster?

Set 4: Cross-Chain Protocols & Bridging Dynamics

- How do cross-chain bridge smart contracts log deposit, lock, burn, and release events across non-interoperable networks?
- What methods reliably correlate an asset locked on Chain A with a corresponding mint/claim on Chain B without central order book access?

- How do cross-chain liquidity networks (e.g., Thorchain, Stargate) obscure deterministic transaction paths compared to wrapped-token bridges?
- What indexing architectures trace simultaneous fund splits across three or more distinct blockchain protocols?
- How can atomic swaps and decentralized bridge relayers be mapped when destination addresses use differing cryptographic formats?

Set 5: Privacy Protocols, Mixers & Obfuscation

- What cryptographic and structural traces remain visible on-chain when funds pass through zero-knowledge mixing pools (e.g., Tornado Cash)?
- How can timing analysis and fixed-denomination deposit/withdrawal matching de-anonymize pool-based mixing services?
- What state changes occur when funds interact with privacy-focused Layer-2 rollups or zero-knowledge protocols?
- How do token-swapping decentralized exchanges (DEXs) and automated market makers (AMMs) act as pseudo-mixers during money laundering?
- What indicators differentiate legitimate DeFi arbitrage bot transactions from deliberate algorithmic obfuscation?

Set 6: Real-Time Ingestion, Mempool & Performance

- What latency bottlenecks arise when executing recursive multi-hop queries on live blockchain RPC nodes, and how can local indexing eliminate them?
- How can mempool-sniffing techniques detect an incoming transfer to an exchange deposit wallet before block finalization?
- How do real-time WebSocket listeners manage chain reorganizations (re-orgs) without corrupting live investigation trees?
- What caching strategies enable sub-second re-traversal of frequently hit intermediary laundering hubs and bridge routers?
- How should a real-time ingestion pipeline handle sudden rate-limiting and node failover across multiple public and private RPC providers?

Set 7: VASP Infrastructure & Deposit Identification

- How do centralized exchange architectures internally route funds from unique user deposit addresses into central hot/cold wallets?
- What sweeping patterns (batching frequency, gas price thresholds, balance triggers) identify an unlabelled wallet as an exchange deposit proxy?
- How can automated scraping of public registries, proofs of reserves, and smart contract tags maintain an up-to-date VASP directory?
- What structural differences separate a direct exchange deposit address from a third-party custodial payment gateway?
- How do P2P escrow smart contracts handle transaction handshakes compared to direct spot-trading deposit addresses?

Set 8: Risk Scoring, Typologies & Machine Learning

- Which feature vectors (velocity, fan-out ratio, hop count, dormancy duration, balance retention) best train anomaly detection classifiers?
- How can unsupervised clustering algorithms (e.g., DBSCAN, HDBSCAN) group

previously unseen laundering typologies without labelled training sets?
- What loss functions and evaluation metrics prevent false positives when classifying high-frequency trading bots versus criminal peelers?
- How can graph neural networks (GNNs) incorporate temporal features (timestamp deltas) to classify illicit transaction subgraphs?
- What rule-based thresholds are required to calibrate risk severity tiers for law enforcement prioritization?

Set 9: Legal Admissibility, Compliance & Asset Freezing

- What technical evidentiary standards (deterministic replayability, cryptographic hash proofs) ensure automated traces are admissible under procedural laws?
- What exact metadata fields (timestamp, tx hash, hop narrative, VASP UID) must be present in a preservation request for an exchange compliance desk to act immediately?
- What differences exist in legal turnaround and documentation for domestic FIU-registered exchanges versus offshore, non-compliant VASPs?
- How do emergency asset preservation orders (Section 91/102 CrPC & Section 94/107 BNSS) legally bridge the gap between initial reporting and formal seizure?
- What digital chain-of-custody protocols prevent allegations of evidence tampering in automated PDF audit exports?

Set 10: System Interoperability & LEA Platform Integration

- How can an automated tracing pipeline securely integrate with national cybercrime portals (e.g., NCRP, SAHYOG, 1930 Helpline APIs)?
- What data interchange standards (e.g., STIX/TAXII, standardized JSON schemas) allow seamless sharing of crypto threat intelligence between agencies?
- How should role-based access control (RBAC) and end-to-end cryptographic logging be designed to maintain investigator operational security?
- What asynchronous task queuing architectures (e.g., Celery, Redis Streams, Kafka) prevent system bottlenecks during high-volume complaint spikes?
- How can the platform generate automated, interactive visual flowcharts alongside exportable legal briefs for non-technical field officers?